

# Oscillator Model

To understand material's response, it is useful to understand oscillator's model.  
Assume the net charge system is 0 and applied by an external field  $E$



When there is a spring, we call Lorentz

$$m\ddot{x} = -fx - \frac{e^2}{\epsilon} x - qE \Rightarrow \ddot{x} + \frac{1}{\epsilon} x + \omega_0^2 x = -\frac{qE}{m} \quad \omega_0^2 = \frac{f}{m} \quad 2\gamma = \frac{f}{m}$$



When the electron is free, we call Drude model  $f=0$

$$m\ddot{x} = -\frac{e^2}{\epsilon} x - qE \Rightarrow \ddot{x} + \frac{1}{\epsilon} x = -\frac{qE}{m}$$

The force by  $f$  is due to the binding interaction between electrons and protons. The friction  $\gamma$  is due to factors such as electrons scattering, interactions with holes, etc...

Using either FT, one arrives:

$$d(\omega) = -q x(\omega) = -\frac{qE(\omega)}{m} \frac{1}{\omega^2 - \omega_0^2 + i\omega\gamma}$$

With  $P(\omega) = x(\omega) E(\omega)$ , we have the response:

$$x(\omega) = -\frac{Ne^2}{m} \frac{1}{\omega^2 - \omega_0^2 + i\omega\gamma} = -\frac{Ne^2}{m} \frac{1}{2\omega_0\gamma} \left[ \frac{1}{\omega - \omega_0 + i\gamma} - \frac{1}{\omega + \omega_0 + i\gamma} \right] \quad \omega_0 = \sqrt{\omega_0^2 - \gamma^2}$$

This form contains so much information.

Let  $A = \omega - \omega_0$ , we see immediately poles

$$A + i\gamma = 0 \Rightarrow \omega = \omega_0 - i\gamma$$

The single number contains two pieces of physics

$$Re(\omega) = \omega_0 \quad Im(\omega) = -\gamma \Rightarrow e^{-i(\omega_0 - i\gamma)t} = e^{-i\omega_0 t} e^{-\gamma t}$$

an oscillation at  $\omega_0$  with a decay of  $\gamma$

The poles are unachievable. Realistically, we scan " $\omega$ " as the real axis for response while  $\gamma$  deviates from the real axis

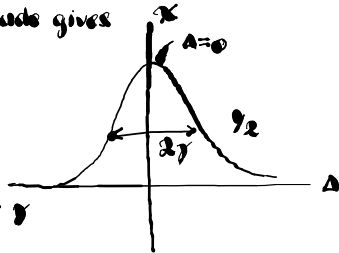
Now, looking at  $x(\omega) = |x(\omega)| e^{i\phi(\omega)}$ , the amplitude gives

$$|x|^2 \propto \frac{1}{A^2 + \gamma^2} \quad \text{Lorentzian}$$

$$\text{At } A=0, |x|^2 \propto \frac{1}{\gamma^2} \Rightarrow \text{maximum}$$

$$\text{At half maximum, } \frac{1}{A^2 + \gamma^2} = \frac{1}{2} \frac{1}{\gamma^2} \Rightarrow |A| = \gamma$$

$$\text{The full width at half maximum } FWHM = 2\gamma = \frac{1}{\tau}$$



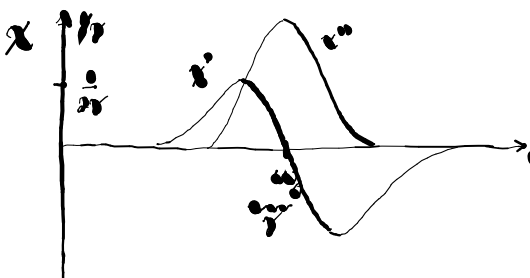
The phase is induced such that the created polarization is out of phase, leading to a bad interference of induced  $E$  field and incoming  $E$  field.

$$\phi(\omega) = -\arctan\left(\frac{\gamma}{A}\right) = \arctan\left(\frac{A}{\gamma}\right)$$

At  $A=0$ , the shift  $\phi(\omega) = -\pi/2 \Rightarrow$  exactly the cause of delay.

The same form can be seen in a different form

$$\frac{1}{A + i\gamma} = \frac{x'}{A^2 + \gamma^2} - \frac{i\gamma}{A^2 + \gamma^2}$$



We call  $x'$  as dispersion and

$x''$  as absorption, but one needs care.

It is wrong to say  $x'$  - phase only and  $x''$  - amplitude.

$$\text{Take } E(t) = E_0 \cos(\omega t)$$

$$P(\omega) = x(\omega) E(\omega)$$

$$= E_0 [x' \cos \omega t + x'' \sin \omega t]$$

The  $x'$  is in phase with  $E(t)$ ,  $x''$  is out of phase

$x'$  describes how materials absorb and

release energy. In one cycle, no net gain.

But, it reveals  $\epsilon_2 \rightarrow$   $\epsilon_2$  is amplitude.

The term  $x''$  removes energy from the field.  $E \frac{dP}{dt}$

$$E \frac{dP}{dt} = E_0^2 \omega [-x' \cos^2 \omega t + x'' \sin^2 \omega t]$$

These are real reasons not sth as  $e^{i\omega t} e^{-\gamma t}$

\* When  $\omega \neq \omega_0$ , a large absorption occurs